

Problem Set 1:

Submission: no submission for Problem Set 1; Solutions: Mon, Apr 20, 2026

Problem 1 Vector Calculus

Let $\mathbf{v}, \mathbf{w}, \mathbf{g}$ be smooth vector fields, let f be a smooth real valued function and let $\mathbf{r}$ denote the position vector $\mathbf{r} = (x, y, z)^T$. Prove the following identities using the totally antisymmetric Levi-Civita symbol ϵ_{ijk} :

(1.a) $\nabla \cdot (f\mathbf{g}) = f \nabla \cdot \mathbf{g} + \mathbf{g} \cdot \nabla f$

(1.b) $\nabla \times (f\mathbf{g}) = f \nabla \times \mathbf{g} - \mathbf{g} \times \nabla f$

(1.c) $\nabla \times (\mathbf{g} \times \mathbf{r}) = 2\mathbf{g} + r \frac{\partial \mathbf{g}}{\partial r} - \mathbf{r} (\nabla \cdot \mathbf{g})$

(1.d) $(\mathbf{v} \times \nabla) \times \mathbf{w} - \mathbf{v} \times (\nabla \times \mathbf{w}) = (\mathbf{v} \cdot \nabla) \mathbf{w} - \mathbf{v} (\nabla \cdot \mathbf{w})$

Now, let $\mathbf{w} = \nabla \varphi$ be a gradient field and $\nabla \cdot \mathbf{v} = 0$, with both $\varphi(\mathbf{r})$ and $|\mathbf{v}(\mathbf{r})|$ decaying faster than $1/r$ as $r \rightarrow \infty$.

(1.e) Use part (a) and Gauß's theorem to show that $\int_{\mathbb{R}^3} d^3\mathbf{r} \mathbf{w} \cdot \mathbf{v} = 0$.

Finally, let $\mathbf{w}$ be a gradient field as before, but now $\nabla \times \mathbf{v} = 0$.

(1.f) Use part (b) and Gauß's theorem to show that $\int_{\mathbb{R}^3} d^3\mathbf{r} \mathbf{w} \times \mathbf{v} = 0$.

Hint: For the identities c and d, the relation

$$\epsilon_{ijk}\epsilon_{nmk} = \delta_{in}\delta_{jm} - \delta_{im}\delta_{jn}, \quad (1)$$

with δ_{ij} denoting the Kronecker-Delta, might be useful.

Solution:

(1.a) $\nabla \cdot (f\mathbf{g}) = f \nabla \cdot \mathbf{g} + \mathbf{g} \cdot \nabla f$

$$\begin{aligned} \nabla \cdot (f\mathbf{g}) &= (\partial_i \mathbf{e}_i) \cdot (f g_j \mathbf{e}_j) \\ &= \partial_i (f g_j) (\mathbf{e}_i \cdot \mathbf{e}_j) \\ &= \partial_i (f g_j) \delta_{ij} \\ &= \partial_i (f g_i) \\ &= (\partial_i f) g_i + f \partial_i g_i \\ &= g_i \partial_i f + f \partial_i g_i \\ &= \mathbf{g} \cdot \nabla f + f \nabla \cdot \mathbf{g}. \end{aligned}$$

$$(1.b) \quad \nabla \times (f\mathbf{g}) = f \nabla \times \mathbf{g} - \mathbf{g} \times \nabla f$$

$$\begin{aligned} \nabla \times (f\mathbf{g}) &= (\partial_j \mathbf{e}_j) \times (f g_k \mathbf{e}_k) \\ &= \partial_j (f g_k) (\mathbf{e}_j \times \mathbf{e}_k) \\ &= \partial_j (f g_k) \varepsilon_{jki} \mathbf{e}_i \\ &= \partial_j (f g_k) \varepsilon_{ijk} \mathbf{e}_i \\ &= (\partial_j f) g_k \varepsilon_{ijk} \mathbf{e}_i + f (\partial_j g_k) \varepsilon_{ijk} \mathbf{e}_i \\ &= -g_k (\partial_j f) \varepsilon_{kji} \mathbf{e}_i + f (\partial_j g_k) \varepsilon_{jki} \mathbf{e}_i \\ &= -\mathbf{g} \times \nabla f + f \nabla \times \mathbf{g}. \end{aligned}$$

$$(1.c) \quad \nabla \times (\mathbf{g} \times \mathbf{r}) = 2\mathbf{g} + r \frac{\partial \mathbf{g}}{\partial r} - \mathbf{r} (\nabla \cdot \mathbf{g})$$

$$\begin{aligned} \nabla \times (\mathbf{g} \times \mathbf{r}) &= (\partial_i \mathbf{e}_i) \times (g_j r_k \varepsilon_{jkl} \mathbf{e}_l) \\ &= \partial_i (g_j r_k) \varepsilon_{jkl} (\mathbf{e}_i \times \mathbf{e}_l) \\ &= \partial_i (g_j r_k) \varepsilon_{jkl} \varepsilon_{iln} \mathbf{e}_n \\ &= \partial_i (g_j r_k) (\delta_{jn} \delta_{ki} - \delta_{ji} \delta_{kn}) \mathbf{e}_n \\ &= \partial_i (g_n r_i) \mathbf{e}_n - \partial_i (g_i r_n) \mathbf{e}_n \\ &= ((\partial_i g_n) r_i + g_n \partial_i r_i) \mathbf{e}_n - ((\partial_i g_i) r_n + g_i \partial_i r_n) \mathbf{e}_n \\ &= (r_i \partial_i g_n + 3g_n - r_n \partial_i g_i - g_n) \mathbf{e}_n \\ &= (2g_n + r_i \partial_i g_n - r_n \partial_i g_i) \mathbf{e}_n \\ &= 2\mathbf{g} + (\mathbf{r} \cdot \nabla) \mathbf{g} - \mathbf{r} (\nabla \cdot \mathbf{g}) \\ &= 2\mathbf{g} + r \frac{\partial \mathbf{g}}{\partial r} - \mathbf{r} (\nabla \cdot \mathbf{g}). \end{aligned}$$

To justify the last step, use the chain rule to write $\frac{\partial}{\partial r} = \frac{\partial x_i}{\partial r} \partial_i = \frac{\partial(\hat{\mathbf{r}}_i r)}{\partial r} \partial_i = \frac{\partial r}{\partial r} \hat{\mathbf{r}}_i \partial_i = \hat{\mathbf{r}} \cdot \nabla$, for which we used that the unit vector $\hat{\mathbf{r}}$ only depends on the polar and azimuthal angle in spherical coordinates. Thus, we immediately get $\mathbf{r} \cdot \nabla = r \frac{\partial}{\partial r}$.

$$(1.d) \quad (\mathbf{v} \times \nabla) \times \mathbf{w} - \mathbf{v} \times (\nabla \times \mathbf{w}) = (\mathbf{v} \cdot \nabla) \mathbf{w} - \mathbf{v} (\nabla \cdot \mathbf{w})$$

$$\begin{aligned} (\mathbf{v} \times \nabla) \times \mathbf{w} - \mathbf{v} \times (\nabla \times \mathbf{w}) &= (v_i \partial_j \varepsilon_{ijk} \mathbf{e}_k) \times \mathbf{w} - \mathbf{v} \times (\partial_i w_j \varepsilon_{ijk} \mathbf{e}_k) \\ &= v_i \partial_j \varepsilon_{ijk} w_l \varepsilon_{kln} \mathbf{e}_n - v_l \partial_i w_j \varepsilon_{ijk} \varepsilon_{lkn} \mathbf{e}_n \\ &= v_i \partial_j w_l (\varepsilon_{ijk} \varepsilon_{kln}) \mathbf{e}_n - v_l \partial_i w_j (\varepsilon_{ijk} \varepsilon_{lkn}) \mathbf{e}_n \\ &= v_i \partial_j w_l (\delta_{il} \delta_{jn} - \delta_{in} \delta_{jl}) \mathbf{e}_n - v_l \partial_i w_j (\delta_{in} \delta_{jl} - \delta_{il} \delta_{jn}) \mathbf{e}_n \\ &= v_i \partial_j w_i \mathbf{e}_j - v_i \partial_j w_j \mathbf{e}_i - \cancel{v_j \partial_i w_j \mathbf{e}_i} + \cancel{v_i \partial_i w_j \mathbf{e}_j} \\ &= v_i \partial_i w_j \mathbf{e}_j - v_i \partial_j w_j \mathbf{e}_i \\ &= (\mathbf{v} \cdot \nabla) \mathbf{w} - \mathbf{v} (\nabla \cdot \mathbf{w}). \end{aligned}$$

(1.e) Using part (a), we can write

$$\begin{aligned}\nabla \cdot (\varphi \mathbf{v}) &= \varphi \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \varphi \\ &= \mathbf{v} \cdot \mathbf{w}.\end{aligned}$$

Next, let $B_R = \{\mathbf{x} \in \mathbb{R}^3 : |\mathbf{x}| < R\}$ be an open ball of radius R centered around the origin. Then:

$$\int_{\mathbb{R}^3} d^3\mathbf{r} \mathbf{w} \cdot \mathbf{v} = \lim_{R \rightarrow \infty} \int_{B_R} d^3\mathbf{r} \nabla \cdot (\varphi \mathbf{v}) = \lim_{R \rightarrow \infty} \int_{\partial B_R} dA \hat{\mathbf{r}} \cdot \varphi \mathbf{v}.$$

Therefore:

$$\begin{aligned}\left| \int_{\mathbb{R}^3} d^3\mathbf{r} \mathbf{w} \cdot \mathbf{v} \right| &\leq \lim_{R \rightarrow \infty} \int_{\partial B_R} dA |\hat{\mathbf{r}} \cdot \varphi \mathbf{v}| \\ &\leq \lim_{R \rightarrow \infty} 4\pi R^2 \sup_{\mathbf{r} \in \partial B_R} |\hat{\mathbf{r}} \cdot \varphi \mathbf{v}| \\ &\leq \lim_{R \rightarrow \infty} 4\pi R^2 \sup_{\mathbf{r} \in \partial B_R} |\varphi \mathbf{v}| \\ &= 0.\end{aligned}$$

The last equality follows from the assumption that both $\varphi(\mathbf{r})$ and $|\mathbf{v}(\mathbf{r})|$ decay faster than $1/r$ as $r \rightarrow \infty$, implying that their product decays faster than $1/r^2$ as $r \rightarrow \infty$.

(1.f) Using part (b), we can write

$$\mathbf{w} \times \mathbf{v} = \nabla \times (\varphi \mathbf{v}).$$

Now, fix a component i of the vector field $\varphi \mathbf{v}$. Define a vector field $\mathbf{g}^{(i)}$ by

$$\mathbf{g}_j^{(i)} \equiv \epsilon_{ijk} \varphi \mathbf{v}_k. \quad (2)$$

Then we have

$$\nabla \cdot \mathbf{g}^{(i)} = \partial_j \epsilon_{ijk} \varphi \mathbf{v}_k = (\nabla \times (\varphi \mathbf{v}))_i = (\mathbf{w} \times \mathbf{v})_i. \quad (3)$$

Now let $B_R = \{\mathbf{x} \in \mathbb{R}^3 : |\mathbf{x}| < R\}$. By the divergence theorem,

$$\int_{B_R} d^3\mathbf{r} (\mathbf{w} \times \mathbf{v})_i = \int_{B_R} d^3\mathbf{r} \nabla \cdot \mathbf{g}^{(i)} = \int_{\partial B_R} dA \hat{\mathbf{r}} \cdot \mathbf{g}^{(i)}. \quad (4)$$

Hence,

$$\begin{aligned}\left| \int_{\mathbb{R}^3} d^3\mathbf{r} (\mathbf{w} \times \mathbf{v})_i \right| &\leq \lim_{R \rightarrow \infty} \int_{\partial B_R} dA |\hat{\mathbf{r}} \cdot \mathbf{g}^{(i)}| \\ &\leq \lim_{R \rightarrow \infty} 4\pi R^2 \sup_{\mathbf{r} \in \partial B_R} |\varphi \mathbf{v}| \\ &= 0.\end{aligned}$$

Again, the last equality follows from the assumption that both $\varphi(\mathbf{r})$ and $|\mathbf{v}(\mathbf{r})|$ decay faster than $1/r$ as $r \rightarrow \infty$, implying that their product decays faster than $1/r^2$ as $r \rightarrow \infty$.

Problem 2 Properties of the Fourier Transform

The Dirac- δ distribution can be understood as the $\epsilon \rightarrow 0$ limit of several continuously differentiable approximations $\delta_\epsilon(x)$. One of them is the normal distribution

$$\delta_\epsilon(x) = \mathcal{C}^{-1} \exp\left(-\frac{x^2}{2\epsilon}\right). \quad (5)$$

(2.a) Derive the value of $\mathcal{C}$ for which δ_ϵ is properly normalized such that

$$\int_{\mathbb{R}} dx \delta_\epsilon(x) = 1. \quad (6)$$

Next, we would like to prove the identity

$$\delta(x) = \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{ikx}, \quad (7)$$

meaning that the Fourier transform of a constant function is proportional to the Dirac- δ distribution. We want to obtain this result by regularizing the integral by approximating the constant function with a very flat normal distribution

$$f_\epsilon(k) = e^{-\epsilon k^2/2}. \quad (8)$$

(2.b) Use this strategy to conclude that (7) follows after taking the limit $\epsilon \rightarrow 0$.

When solving differential equations such as the wave equation, a common strategy is to transform the equation in momentum space, where the Green's function ($\rightarrow$ next week's problem set) can often be obtained quite easily. The Fourier transform is formally equivalent to a plane wave decomposition of the desired solution. It is therefore instructive to understand how differential operators such as the Nabla operator act on plane waves. To this end, let

$$\mathbf{A}(\mathbf{r}) = \mathbf{c} e^{i\mathbf{k} \cdot \mathbf{r}},$$

where $\mathbf{c}$ and $\mathbf{k}$ are constant vectors. Show that, in every case, the formal replacement

$$\nabla \longrightarrow i\mathbf{k}$$

produces the correct answer for

(2.c) $\nabla \cdot \mathbf{A}$

(2.d) $\nabla \times \mathbf{A}$

(2.e) $\nabla \times (\nabla \times \mathbf{A})$

(2.f) $\nabla(\nabla \cdot \mathbf{A})$

(2.g) $\nabla^2 \mathbf{A}$

Solution:

(2.a) Using the integral of a Gaussian, we have

$$\int_{\mathbb{R}} dk \exp\left(-\frac{k^2}{2\epsilon}\right) = \sqrt{2\pi\epsilon}, \quad (9)$$

hence

$$\mathcal{C} = \sqrt{2\pi\epsilon}. \quad (10)$$

(2.b) Using the Fourier transform of a Gaussian, we find

$$\int_{-\infty}^{\infty} dk \exp\left(-\frac{\epsilon k^2}{2}\right) e^{ixk} = \sqrt{\frac{2\pi}{\epsilon}} e^{-x^2/(2\epsilon)}, \quad (11)$$

implying that $\delta_{\epsilon}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{ikx} f_{\epsilon}(k)$, as desired.

Before showing the action of the Nabla operator on the vector field $\mathbf{A}$, note that

$$\partial_j e^{i\mathbf{k} \cdot \mathbf{r}} = \partial_j e^{ik_{\ell} x_{\ell}} = ik_j e^{i\mathbf{k} \cdot \mathbf{r}}.$$

Since $\mathbf{c}$ is constant, every derivative acts only on the exponential. In components,

$$A_m(\mathbf{r}) = c_m e^{i\mathbf{k} \cdot \mathbf{r}},$$

so

$$\partial_j A_m = c_m \partial_j e^{i\mathbf{k} \cdot \mathbf{r}} = ik_j c_m e^{i\mathbf{k} \cdot \mathbf{r}}.$$

Thus each occurrence of ∂_j may indeed be replaced by ik_j .

(2.c) **Divergence**

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \partial_j A_j \\ &= \partial_j (c_j e^{i\mathbf{k} \cdot \mathbf{r}}) \\ &= c_j \partial_j e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= ic_j k_j e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= i(\mathbf{k} \cdot \mathbf{c}) e^{i\mathbf{k} \cdot \mathbf{r}}. \end{aligned}$$

This is exactly what we get from the replacement $\nabla \rightarrow i\mathbf{k}$.

(2.d) **Curl**

$$\begin{aligned} (\nabla \times \mathbf{A})_i &= \varepsilon_{ijk} \partial_j A_k \\ &= \varepsilon_{ijk} \partial_j (c_k e^{i\mathbf{k} \cdot \mathbf{r}}) \\ &= \varepsilon_{ijk} c_k \partial_j e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= i\varepsilon_{ijk} k_j c_k e^{i\mathbf{k} \cdot \mathbf{r}}. \end{aligned}$$

Therefore

$$\nabla \times \mathbf{A} = i(\mathbf{k} \times \mathbf{c}) e^{i\mathbf{k} \cdot \mathbf{r}}.$$

Again this agrees with $\nabla \rightarrow i\mathbf{k}$

(2.e) **Double curl**

Since $\mathbf{k} \times \mathbf{c}$ is another constant vector, the result follows by applying our result for the curl of $\mathbf{A}$ twice.

(2.f) **Gradient of the divergence**

From earlier,

$$\nabla \cdot \mathbf{A} = i(\mathbf{k} \cdot \mathbf{c}) e^{i\mathbf{k} \cdot \mathbf{r}},$$

where $\mathbf{k} \cdot \mathbf{c}$ is constant. Hence

$$\begin{aligned}\nabla(\nabla \cdot \mathbf{A}) &= \nabla (i(\mathbf{k} \cdot \mathbf{c}) e^{i\mathbf{k} \cdot \mathbf{r}}) \\ &= i(\mathbf{k} \cdot \mathbf{c}) \nabla e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= i(\mathbf{k} \cdot \mathbf{c})(i\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= -\mathbf{k}(\mathbf{k} \cdot \mathbf{c}) e^{i\mathbf{k} \cdot \mathbf{r}}.\end{aligned}$$

This also matches $\nabla \rightarrow i\mathbf{k}$

(2.g) **Laplacian**

In components,

$$\begin{aligned}(\nabla^2 \mathbf{A})_m &= \partial_j \partial_j A_m \\ &= \partial_j \partial_j (c_m e^{i\mathbf{k} \cdot \mathbf{r}}) \\ &= c_m \partial_j \partial_j e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= c_m (ik_j)(ik_j) e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= -k_j k_j c_m e^{i\mathbf{k} \cdot \mathbf{r}}.\end{aligned}$$

Therefore

$$\nabla^2 \mathbf{A} = -k^2 \mathbf{c} e^{i\mathbf{k} \cdot \mathbf{r}} = -k^2 \mathbf{A}.$$

This is again exactly what the replacement rule gives.